Qatar-2b

R-0975

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-2_250619` · created 2026-07-29T18:28:34+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460845.7069 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.066
Transit depth (Rp/Rs)^2	5.19 +/- 2.99 [%]
Orbital Inclination (inc)	85.87 +/- 1.73
Airmass coefficient 1 (a1)	2.94 +/- 1.77
Airmass coefficient 2 (a2)	-0.69 +/- 0.52
Scatter in the residuals of the lightcurve fit is	57.89 %
Best Comparison Star	#1 - [90, 244]
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.0721 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.228 ± 0.066 vs 0.16327 (deviation 0.98σ)
Duration fit vs published	0.0721 d vs 0.0767 d (deviation 0.79σ)
Mid-time O–C	-0.4 min
Depth SNR	3.5
Residual scatter	57.89 %

Light curve

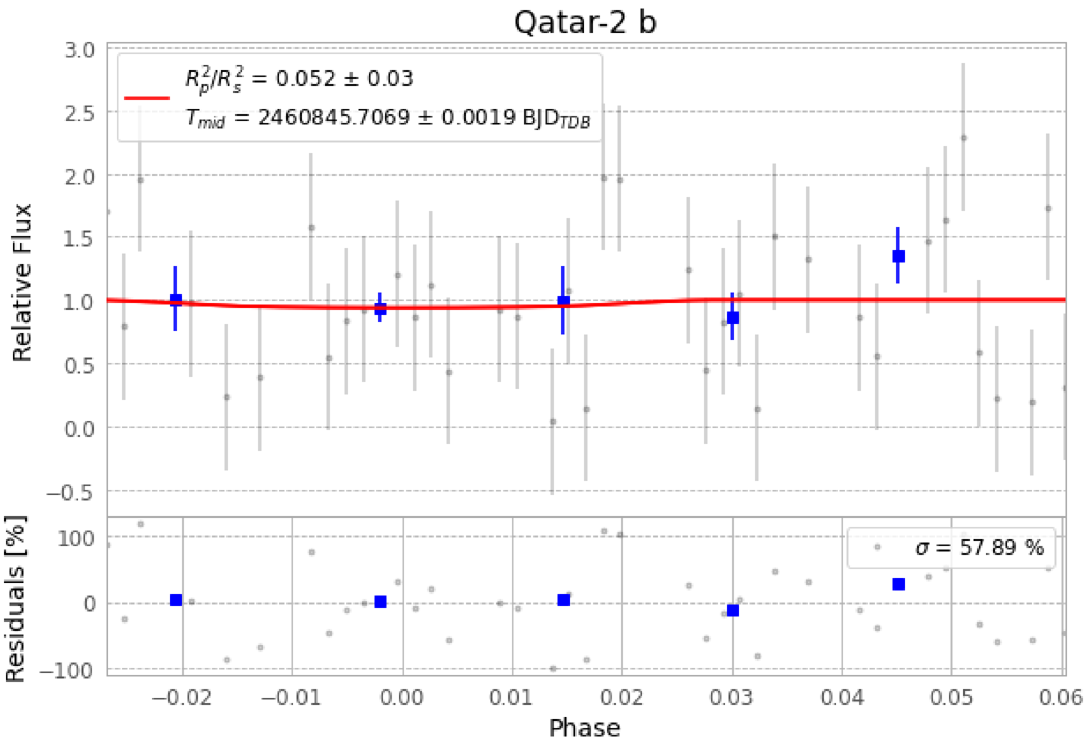

Figure 1 — FinalLightCurve_Qatar-2 b_2025-06-18.png (SHA-256 1148d9f31d5c17a3…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [614, 435], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e0fed13d1239b18c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

57 FITS frames (37 MB), `Qatar-2250619040015.FITS` → `Qatar-2250619064817.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250619.log (afb46658df14…), FinalLightCurve_Qatar-2 b_2025-06-18.png (1148d9f31d5c…), FinalLightCurve_Qatar-2 b_2025-06-18.csv (fa6bae061230…)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)